Learning-Rate Schedules Are Not Adiabatic: A Rate-Dependent Correction for Loss-Curve Prediction

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`https://github.com/wjjpku/DL-final`

Abstract

State-of-the-art laws for predicting the LLM pretraining loss curve under a learning-rate (LR) schedule—the Multi-Power Law (MPL) and the Functional Scaling Law (FSL)—are *adiabatic*: they predict the quasi-static loss a schedule would reach at equilibrium, depending on the schedule only through its integrated history. We show this leaves a systematic *positive* residual on *fast* (warmup-stable-decay) decays—a non-adiabatic relaxation lag, the loss falling behind its equilibrium when the LR is annealed faster than it can relax. Under weak stability assumptions we derive a linear-response law: the leading residual is a causal convolution of per-step LR decrements with a relaxation kernel in cumulative-LR time. The operational correction is its lowest-capacity one-pole closure; in a noise-dominated AdamW specialization the amplitude is the equilibrium-floor sensitivity and the relaxation time is the classical $\tau \sim 1/(\eta\lambda)$. Public LLM curves are consistent with this mechanism ($p = 0.84{\pm}0.17$), while our independently trained small transformer confirms the lag but not a clean inverse-LR scaling. The kernel's rate constant is empirically stable across the public scales, enabling a zero-target-fitting transfer that cuts held-out sharp-decay error by 44%. We then upgrade the law's *amplitude structure* in two ways. First, a *B-identity*: MPL's own annealing coefficient B *is* the equilibrium-floor sensitivity the correction needs, so the amplitude transfers from MPL's published parameters with no additional measurements (-43%, matching the probe-based chain). Second, the public-bed equilibrium floor reads *superlinear* in η ($F_{\mathrm{eq}} \propto \eta^{1+\delta}$, $1{+}\delta = 1.06$–1.49 from three-point fits at $100/400$M)—though we show this reading is protocol-dependent (confound-free equal-S ladders on our own beds give a *sublinear* exponent ≈ 0.65, robust to model size, batch size, and horizon; §6.1), so the superlinearity is a property of the settled-probe protocol, not established as a bed-independent law. The corresponding second-order amplitude closure $\chi(\eta) \propto \eta^{\delta}$ with default $\delta{=}1/4$ (Pareto-selected on the audited matrix, never fit on targets) improves that transfer matrix on every aggregate and lifts cheap-probe calibration by 35–70% relative. An independent reproduction—a separately-trained $\sim$10M transformer and a from-scratch AdamW simulation—confirms the effect (the fast sweep ends higher *despite* 37% more cumulative LR), the amplitude identity (within 3%), and, on never-scanned out-of-family schedules, the second-order closure's cheap-probe gains. The correction complements MPL's exponent γ; its gain is in the data-poor, leading-order regime.

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1 Introduction

The learning-rate schedule shapes the *entire* loss trajectory of LLM pretraining [14, 15, 16]. *Schedule-aware loss-curve prediction* asks: given $\{\eta_t\}$, predict $L(t)$, so that schedules can be compared and optimized *before* spending compute. The defining test is cross-schedule generalization—fit on cosine schedules, predict the unseen WSD family. The state of the art is the **Multi-Power Law** (MPL) [2], with a matching kernel-regression theory, the **Functional Scaling Law** (FSL) [3]:

$$L(t) = L_0 + A\,S(t)^{-\alpha} + B\sum_{k\le t}\Delta\eta_k\, G\big(\eta_k^{-\gamma}(S(t)-S(k))\big), \quad G(x) = 1-(1+Cx)^{-\beta}, \tag{1}$$

where $S(t) = \sum_{i\le t}\eta_i$ and $\Delta\eta_k = \eta_{k-1} - \eta_k$.

The gap: existing laws are adiabatic. These laws *do* use the LR decrements: MPL convolves $\Delta\eta_k$ with a saturating kernel G, and its factor $\eta_k^{-\gamma}$ is a rate-of-decay knob. But this single empirical exponent is their *only* handle on how *fast* the LR changes. Structurally they predict the *quasi-static* loss—the value the schedule would reach if the loss stayed at equilibrium; the river-valley analysis [4] makes this explicit, modelling the noise penalty as set *instantaneously* by the current LR. We call such laws *adiabatic*. Calibrated on slow schedules, γ does not extrapolate to fast WSD decays, where the loss genuinely lags equilibrium—leaving a structured residual (Fig. 2) that no single rate exponent captures across schedules. Our correction is a derived transient *complementary* to γ, which tunes the adiabatic kernel.

Contributions.

- We show that adiabatic loss-curve laws leave a systematic *positive* residual on fast decays, and identify it as a non-adiabatic relaxation lag (§3).
- Under weak moving-equilibrium assumptions we derive a linear-response law for the non-adiabatic residual, and use its lowest-capacity one-pole closure as the rate-dependent term $\mathrm{DropRelaxS}$ (§4). A noise-dominated AdamW specialization identifies the amplitude with $dL_{\mathrm{eq}}/d\eta$ and gives the classical $\tau \sim 1/(\eta\lambda)$ relaxation time; our point is that this timescale governs the *loss-curve-law residual*.
- Public LLM curves show residual relaxation consistent with $\tau \propto 1/\eta$ ($p = 0.84 \pm 0.17$), while the separately trained ∼10M reproduction confirms the lag but does not resolve this scaling; we therefore use inverse-LR relaxation as a mechanism check, not as the main empirical claim (§5, §6).
- The kernel's rate constant is empirically stable on the public same-architecture scales, enabling a low-calibration cross-scale transfer that improves held-out sharp-decay prediction error (MAE) by 44% with no target-curve fitting (§5).
- We prove a *B-identity*: MPL's fitted annealing coefficient B equals the MPL-implied equilibrium-floor sensitivity $dL_{\mathrm{eq}}/d\eta$, and verify it against independent floor measurements (5–15% at all scales). The correction's amplitude then transfers from MPL's own published parameters with no additional measurements (-43% on the same protocol; §5).
- The public-bed equilibrium floor reads *superlinear* in η at the two larger scales (three-point fits, qualitative; marginal at 25M); we use it to derive and evaluate the corresponding second-order amplitude closure $\chi(\eta) \propto \eta^{\delta}$: with the Pareto-selected default δ=1/4 it improves the audited transfer matrix on every aggregate (the hardest cell, cosine→WSD, nearly doubles its gain) and strengthens cheap-probe calibration by 35% relative (~ 65–70% for the in-sample δ=1/2 variant); we also report exactly which part of that gain is amplitude recalibration rather than shape (§5). The naive NQM route to $\chi(\eta)$ is falsified at magnitude; a bulk+edge spectral mechanism is consistent with all observations (§4).
- A second pre-registered, adversarially-audited round (§6.1) sharpens the scope: confound-free equal-S floor ladders read *sublinear* ($\approx$ 0.65, robust across 10.7/25M, batch size, and horizon to 24k), so the public superlinearity is shown to be settled-probe-*protocol*-dependent rather than a parameter-count law; the post-drop slow-mode clock is batch-size-*blind* (not optimizer-noise relaxation); and direct Hessian spectroscopy closes the spectral edge-re-pinning account three ways, leaving the slow mode phenomenological on this bed.

- We give a fair-baseline diagnostic separating a genuine formula gain from an information gap, clarifying exactly when the correction helps, and close two structural dead ends (multi-pole/Lomax kernel shapes; cross-family rate unification) (§5, §2).

2 Related work

Loss-curve laws. Tissue et al. [1], MPL [2], and FSL [3] predict the trajectory from integrated-LR statistics (cumulative LR / intrinsic time / convolutional functionals). FSL and its follow-ups also derive optimal schedules [3]. They are adiabatic in our operational sense: after fitting the quasi-static schedule response, they do not explicitly model the transient tracking error caused by a moving LR-dependent noise floor. Our rate-dependent term is a low-capacity linear-response correction to that residual and is complementary to MPL's exponent γ (which tunes the adiabatic kernel), not a replacement for it.

Cooldown dynamics and schedule geometry. The river-valley picture [4] explains qualitatively why decay "reveals progress", and explicitly models the noise penalty as *instantaneous* in the current LR—the adiabatic assumption we refine. Dremov et al. [5] study the cooldown empirically (including the role of AdamW's β_2 and the decay shape), framing it through a bias–variance trade-off rather than a rate-dependent lag. Late-decay analyses [14] and schedule-shape searches [13] study related questions (why a long high-LR phase helps; what optimal shapes look like) but do not model the loss's rate dependence.

SGD relaxation physics. Our weak-assumption derivation follows the stochastic-approximation idea that local stability of a fixed-LR system yields tracking of slowly moving equilibria [6, 7]; constant-step SGD can also be viewed through an approximate stationary distribution [8]. That a quadratic mode relaxes with $\tau \sim 1/(\eta\lambda)$ and that the SGD noise floor scales with η are classical, from noisy-quadratic and Langevin analyses of SGD [9]; adaptive methods inherit a preconditioned version [10], operating at an adaptive edge of stability [11, 12]. We do not claim these ingredients as new. Our contribution is the synthesis: identifying that adiabatic loss-curve laws miss this moving-equilibrium relaxation lag, deriving the causal decrement-convolution structure, and building a low-capacity cross-scale predictor from it.

3 The non-adiabatic residual

We take a *well-fit* MPL (the published parameters [2]) and examine its prediction residual $r(t) = L_{\mathrm{true}}(t) - L_{\mathrm{MPL}}(t)$ on held-out schedules (Fig. 1). (Isolating the non-adiabatic term requires a well-fit MPL: a baseline that already captures the *adiabatic* part cleanly, so the remainder is the lag and not a backbone mis-fit; a cosine-only fit conflates the two and gives a far noisier residual.) The residual has a sharp signature (Fig. 2): it is ≈ 0 throughout the long stable phase and on gradual cosine decay, but jumps *positive* exactly when the LR decays steeply, and is largest—and *undamped at the curve's end*—for the sharpest decays (`wsd`, `wsdld`; the effect is visible at all three scales, Fig. 3). The end-of-curve lag is $+(6\text{–}15) \times 10^{-3}$ on WSD/WSDLD versus $\lesssim 4 \times 10^{-3}$ on cosine. Crucially, a gradual cosine and a sharp WSD decay reach the *same* final LR, yet only the fast one lags: same destination, faster sweep, bigger lag—the textbook signature of a rate-dependent (non-adiabatic) effect, and exactly the dependence an adiabatic law cannot represent.

4 Theory: a linear-response correction

The empirical signature in §3 is a tracking failure: after a sharp LR drop, the quasi-static noise floor decreases immediately, but the optimizer's excess-loss state relaxes only over a finite time. We make this statement precise in two layers. The main text states the weak-assumption linear-response result and the low-capacity closure used for prediction; the detailed proof is deferred to Appendix A. AdamW is used only as a mechanistic specialization that identifies the amplitude and the $\tau \propto 1/\eta$ timescale, not as a necessary assumption for the convolutional form. This style of argument follows the standard stochastic-approximation view that constant- or slowly-varying-step optimizers track stable local equilibria [6, 7], and is consistent with the stationary-distribution view of constant-step SGD [8].

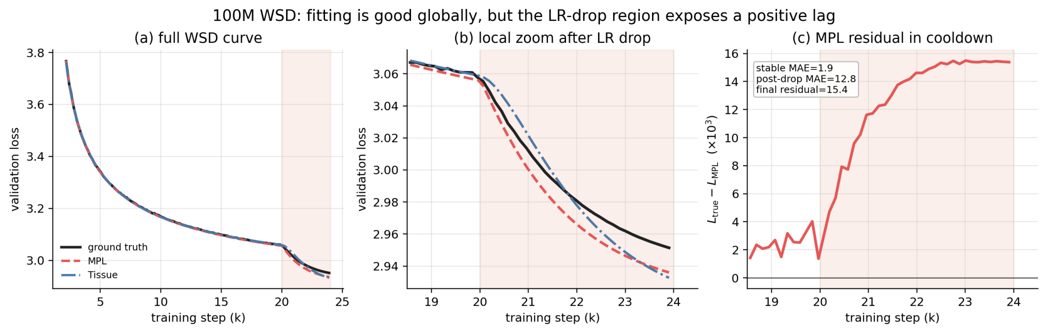

Figure 1: Curve-level reproduction and local WSD fit at 100M. The reproduced Tissue and MPL predictors fit the global trajectory well, but the cooldown zoom reveals that MPL systematically underestimates the true loss immediately after the sharp LR drop. The residual panel isolates the non-adiabatic lag used by the correction.

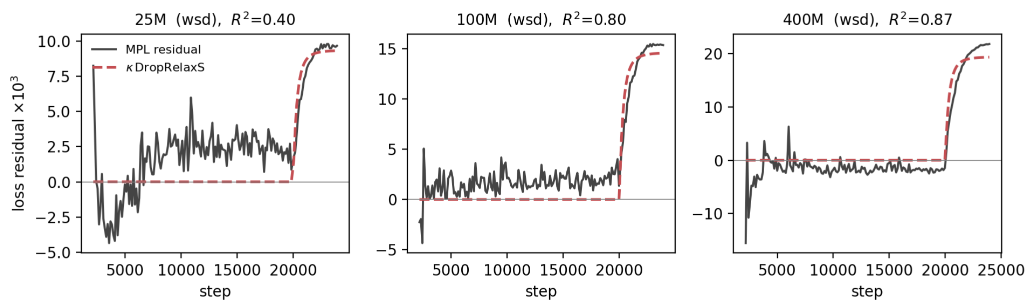

Figure 2: The well-fit MPL prediction residual on the held-out `wsd` curve (solid) is ≈ 0 during the stable phase and jumps up at the step-20000 decay. The derived rate-dependent term $\kappa\,\mathrm{DropRelaxS}$ (dashed) captures it (R^2=$0.40/0.80/0.87$ at $25/100/400$M; the signal, and the fit, grow with model size).

Adiabatic component and hidden excess-loss state. Let

$$S_t = \sum_{u \le t} \eta_u$$

be cumulative LR. We decompose the loss as

$$L_t = L_{\mathrm{ad}}(S_t, \eta_t) + r_t, \tag{2}$$

where L_{ad} denotes the quasi-static component that would be obtained if the LR-dependent excess-loss state were equilibrated at the current LR. In experiments, a well-fit MPL is used as an empirical surrogate for L_{ad}; hence $L_t - L_{\mathrm{MPL},t}$ estimates r_t up to the remaining MPL approximation error.

The weak modeling assumption is that the LR-dependent excess loss is controlled by a hidden state z_t obeying

$$z_{t+1} = \Phi_{S_t,\eta_t}(z_t), \tag{3}$$

and that, for fixed (S, η) in the decay window, there is a locally stable equilibrium $z^\star(S, \eta)$ satisfying

$$z^\star(S, \eta) = \Phi_{S,\eta}\big(z^\star(S, \eta)\big).$$

The scalar excess-loss observable is $F_t = \ell(z_t)$, with equilibrium value

$$F_{\mathrm{eq}}(S, \eta) = \ell\big(z^\star(S, \eta)\big).$$

This assumption is intentionally weaker than a quadratic-loss or diagonal-noise model: z_t may be a covariance state, a preconditioned noise-floor state, or any locally stable state whose equilibrium depends on the LR.

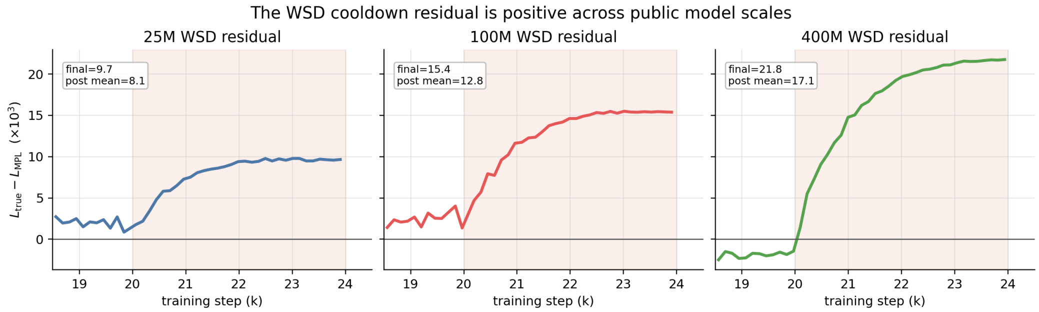

Figure 3: WSD cooldown residuals around the LR drop for all three public model scales. The positive post-drop residual is visible at $25/100/400$M, so the effect is not a single-curve artifact.

Weak assumptions used by the main result. The detailed proof in Appendix A uses only local conditions on the decay window:

(A1) *Smooth moving equilibrium.* $\Phi_{S,\eta}$, $z^\star(S,\eta)$, and ℓ are twice differentiable near the observed trajectory.

(A2) *Local stability.* The linearized maps $T_t = D_z\Phi_{S_t,\eta_t}(z^\star(S_t,\eta_t))$ have uniformly summable products: $\|T_{t-1}\cdots T_{k+1}\| \le q_{k,t}$ with $\sup_t \sum_{k<t} q_{k,t} < \infty$.

(A3) *Separated slow drift.* The first-order response to the gradual change of S_t is already included in the adiabatic backbone up to a small nuisance ϵ_t^S; we isolate the response forced by LR decrements.

(A4) *First-order optimizer time.* On the active subspace, $T_t = I - \eta_t A + E_t$ with stable bounded A and $\|E_t\| \le C(\eta_t^2 + \epsilon_{\mathrm{drift}})$ over the cooldown window.

(A5) *Positive response cone, only for the sign claim.* $\partial_\eta z^\star$ lies in a cone preserved by T_t, and $D\ell$ is nonnegative on that cone.

A1–A4 are enough to derive the current law's convolutional form; A5 is only a sufficient condition for the positive residual sign.

Proposition: causal LR-decrement response. Define the tracking error

$$\delta_t = z_t - z^\star(S_t,\eta_t), \qquad \Delta\eta_t^+ = (\eta_t - \eta_{t+1})_+.$$

(Whether the per-decrement sensitivity is evaluated at the pre- or post-decrement LR is an $O(\Delta\eta^2)$ choice within the first-order expansion; the implementation and Eq. (15) below use the post-decrement convention, matching Eq. (1).) Under the above local assumptions, the cooldown-specific residual has the first-order form

$$r_t = D\ell_t\big[T_{t-1}\cdots T_0\delta_0\big] + \sum_{k<t} K_{k,t}\Delta\eta_k^+ + \mathcal{E}_t, \tag{4}$$

where $D\ell_t = D\ell_{z^\star(S_t,\eta_t)}$ and

$$K_{k,t} = D\ell_t[T_{t-1}\cdots T_{k+1}\,\partial_\eta z^\star(S_k,\eta_k)]\,. \tag{5}$$

The first term is an initial transient; it is small when the decay begins after a long stable phase. The error $\mathcal{E}_t$ is second order in the perturbation size and contains the neglected S-drift and MPL-surrogate error. A representative bound is given in Appendix A; schematically,

$$|\mathcal{E}_t| \lesssim \sum_{k<t} q_{k,t}\big(|\Delta\eta_k|^2 + \|\delta_k\|^2 + \|\epsilon_k^S\| + \eta_k^2\big) + \epsilon_{\mathrm{MPL}}, \tag{6}$$

where $q_{k,t}$ is the stable propagation envelope and ϵ_k^S is the slow-geometry forcing. Thus the leading non-adiabatic residual is not an arbitrary bump: it is the causal linear response of a stable moving equilibrium to LR decrements.

Why the memory is in cumulative-LR time. To obtain the intrinsic-time kernel, add the standard first-order optimizer approximation

$$T_t = I - \eta_t A + O(\eta_t^2 + \epsilon_{\mathrm{drift}}), \tag{7}$$

on the finite-dimensional active subspace that controls the excess loss, with bounded stable operator A *treated as approximately constant over the decay window* (this is the static-spectrum idealization; its slow variation is carried by $\epsilon_{\mathrm{drift}}$, and constancy is precisely what turns the product into a genuine S-time exponential rather than a time-ordered one). Then

$$T_{t-1} \cdots T_{k+1} = \exp[-A(S_t - S_k)] + O\left(\sum_{u=k+1}^{t-1} \eta_u^2 + \epsilon_{\mathrm{drift}}\right). \tag{8}$$

If the response of A under the observable $D\ell$ admits a positive Laplace-mixture representation (equivalently, K is completely monotone—which holds when A is symmetric positive and the forcing $\partial_\eta z^\star$ aligns positively with $D\ell$, as for a noise-floor relaxation), then

$$r_t \approx \sum_{k<t} \underbrace{\int e^{-a(S_t - S_k)}\, d\rho(a)}_{K(S_t - S_k)} \Delta\eta_k^+. \tag{9}$$

This is the primitive scaling law: a fast-cooldown residual is a convolution of LR decrements with a stable memory kernel in cumulative-LR time. Its instantaneous amplitude is fixed by the equilibrium-floor LR sensitivity at fixed progress:

$$K(0) = D\ell_{z^\star} \partial_\eta z^\star = \partial_\eta F_{\mathrm{eq}}(S, \eta)\big|_S =: \chi(S, \eta). \tag{10}$$

One-pole closure and the operational law. The general kernel K in Eq. (9) is not used as a flexible fit. To avoid overfitting, we use the lowest-capacity stable closure: a single pole in S-time, with a floor-sensitivity amplitude estimated from independent equilibrium probes. Approximating $\chi(S, \eta)$ by a constant χ over the decay window gives

$$\boxed{L(t) = L_{\mathrm{MPL}}(t) + c\,\chi \underbrace{\sum_{k<t} e^{-\lambda_{\mathrm{slow}}(S_t - S_k)}(\eta_{k-1} - \eta_k)_+}_{\mathrm{DropRelaxS}_{\lambda_{\mathrm{slow}}}(t)}} \tag{11}$$

(LR-drop relaxation in S-time; the decrement into η_k is deposited at S_k), where $c \le 1$ accounts for compressing a spectral mixture into one effective pole and for residual mismatch between MPL and the ideal adiabatic component. If the implementation defines $\mathrm{DropRelaxS}$ using normalized drops $(\eta_{k-1} - \eta_k)_+/\eta_{\mathrm{peak}}$, then the prefactor is equivalently $c\eta_{\mathrm{peak}}\chi$; with raw LR drops, the dimensionally invariant amplitude is $c\chi$.

Corollary: current law under the weak assumptions. Under A1–A4, if the initial transient at the decay start is negligible and the scalar response kernel is dominated by one slow mode $a = \lambda_{\mathrm{slow}}$, then the leading cooldown residual is exactly Eq. (11) up to the second-order remainder in Eq. (6). Thus the operational correction does not require a quadratic network model: it is the one-pole closure of a stable moving-equilibrium response. The AdamW calculation below is only used to identify χ and to motivate the expected scale of λ_{slow}.

AdamW specialization. The preceding response law does not rely on AdamW. AdamW supplies a local effective-mode calculation that explains the observed amplitude and timescale. This reduction ignores momentum transients, bias correction, decoupled weight decay, the numerical ϵ, and the finite-time dynamics of v_t; it should be read as a local noisy-quadratic approximation. For a Hessian mode with curvature λ_i and gradient-noise standard deviation s_i, noise-dominated AdamW has $v_i^\star \approx s_i^2$, so

$$x_{i,t+1} = \left(1 - \eta_t \frac{\lambda_i}{s_i}\right) x_{i,t} - \eta_t \frac{1}{s_i} \xi_{i,t}, \qquad \mathbb{E}\xi_{i,t}^2 = s_i^2.$$

For $V_{i,t} = \mathbb{E}[x_{i,t}^2]$ and fixed LR η,

$$V_{i,t+1} = \left(1 - \eta \frac{\lambda_i}{s_i}\right)^2 V_{i,t} + \eta^2, \tag{12}$$

so

$$V_i^\star(\eta) = \frac{\eta^2}{1-(1-\eta\lambda_i/s_i)^2} = \frac{\eta s_i}{2\lambda_i} + O(\eta^2).$$

The modal excess-loss floor is

$$F_i^\star(\eta) = \frac{1}{2}\lambda_i V_i^\star(\eta) = \frac{\eta s_i}{4} + O(\eta^2), \qquad \frac{dF_i^\star}{d\eta} = \frac{s_i}{4} + O(\eta),$$

and perturbations decay as

$$\left(1-\eta\frac{\lambda_i}{s_i}\right)^2 = \exp\left[-2\eta\frac{\lambda_i}{s_i} + O(\eta^2)\right].$$

Thus the AdamW specialization of Eq. (9) is the spectral mixture

$$\Delta L(t) \approx \sum_i \frac{s_i}{4}\sum_{k<t} e^{-2(\lambda_i/s_i)(S_t-S_k)}(\eta_k-\eta_{k+1})_+, \qquad \sum_i \frac{s_i}{4} = \frac{dF_{\rm eq}}{d\eta} + O(\eta). \tag{13}$$

It predicts a positive lag, an amplitude given by the equilibrium-floor sensitivity, and a step-time relaxation rate

$$\tau_i^{-1} = 2\eta\frac{\lambda_i}{s_i}, \qquad \tau_i \propto \eta^{-1}.$$

The classical $\tau \sim 1/(\eta\lambda)$ timescale is not itself new; our claim is that this relaxation controls the residual left by adiabatic loss-curve laws after fast LR decay.

The B-identity: MPL already contains the amplitude. The chain above needs the floor sensitivity $\chi = dL_{\rm eq}/d\eta$, which we measured from two-stage probes. It is in fact already encoded in MPL. Compare two schedules that agree up to step k, after which one holds η and the other drops to $\eta - d\eta$ and holds. MPL's annealing term changes by $-B\,d\eta\,G(\eta^{-\gamma}(S_t-S_k))$ and $G \to 1$ as $S_t - S_k \to \infty$, while the $A\,S^{-\alpha}$ backbone difference vanishes at matched S. Hence the *equilibrated* loss change per unit LR drop is exactly B:

$$\left.\frac{\partial L_{\rm eq}^{\rm MPL}}{\partial\eta}\right|_S = B, \tag{14}$$

i.e. MPL structurally encodes a *linear* equilibrium floor with slope B. This identity is falsifiable—B is fit to whole loss curves, the floor slope is measured from settled two-stage finals—and it holds: $B = 363.8/437.9/523.4$ versus measured $dL_{\rm eq}/d\eta = 381.7/499.3/561.5$ at 25/100/400M (5–15%, same $\sim N^{0.13}$ trend; equal at fixed S to $\partial_\eta F_{\rm eq}$ since the backbone drops out). Consequence: the correction's amplitude can be predicted from MPL's *own* parameters, $\kappa = c''\eta_{\rm peak}B$, with no additional measurements (§5).

Second-order closure: the amplitude is η-dependent. Eq. (11) froze $\chi(S,\eta)$ to a constant. The derivation itself says more: the forcing in Eq. (5) is $\partial_\eta z^\star(S_k,\eta_k)$, evaluated at the LR *where the decrement is deposited*. The settled floors are measurably *superlinear*, $F_{\rm eq} \propto \eta^p$ with $p = 1.06/1.49/1.25$ at 25/100/400M (three-point fits; qualitative $p > 1$), so $\chi(\eta) \propto \eta^{p-1}$ falls by up to $\sim 3\times$ over a decade of decay at 100/400M (negligibly at 25M). The second-order one-pole closure weights each decrement by the post-decrement LR:

$$L(t) = L_{\rm MPL}(t) + \kappa\sum_{k<t}\left(\tfrac{\eta_k}{\eta_{\rm peak}}\right)^\delta e^{-\lambda_{\rm slow}(S_t-S_k)}\,\tfrac{(\eta_{k-1}-\eta_k)_+}{\eta_{\rm peak}}, \qquad \delta = (p-1)_+, \tag{15}$$

with $\delta = 1/4$ as the default, selected by Pareto dominance on the audited transfer matrix (§5), never fit on target curves; the measured per-scale $(p-1)_+$ behaves equivalently, and we do not claim universality of either choice beyond the audited curves. Eq. (11) is the δ=0 special case (with $\kappa = c\,\eta_{\rm peak}\chi$ under the normalized-drop convention of §6).

Where the superlinearity cannot and can come from. It cannot come from the same slow noise-dominated modes that set λ_{slow}: the exact mode response gives $\chi_i(\eta) = s_i/(2-\eta\lambda_i/s_i)^2 = s_i/4 + (\lambda_i/4)\eta + O(\eta^2)$, and with the measured $\eta_{\text{peak}}\lambda_{\text{eff}} \sim 2\times10^{-3}$ this varies by $\sim 0.2\%$ across the decay—three orders of magnitude short. A mechanism consistent with all observations is a two-population spectrum: a *bulk* of slow modes contributing the linear floor $\propto \sum_i s_i\eta/4$, plus an *edge* tail of preconditioned rates, $g(a) \propto a^{-\zeta}$, truncated at the adaptive stability boundary $a_{\max} \approx 2\theta/\eta$ *at the current LR* [12]. Substituting $u = \eta a$ in the edge integral gives

$$F_{\text{edge}}(\eta) = g_0\,\eta^{\zeta}\int_{u_{\min}}^{2\theta}\frac{u^{-\zeta}}{2(2-u)}\,du \;\propto\; \eta^{\zeta}$$

(the lower cutoff $u_{\min} = \eta a_{\min}$ contributes $\propto a_{\min}^{1-\zeta}\eta$, which is absorbed into the bulk linear term), hence

$$F_{\text{eq}}(\eta) = c_1\eta + c_2\eta^{\zeta}, \qquad \zeta \in (1,2), \tag{16}$$

whose one-decade power-law fit yields exactly the observed effective $p \in (1,\zeta)$. A frozen cutoff ($a_{\max}$ pinned at η_{peak}) predicts a near-linear floor instead, so the two are distinguishable. The non-closed step, stated honestly: in a static spectrum the edge population relaxes in $O(1)$ steps and would contribute no slow lag; the conjecture is that the edge *re-pins* to the new boundary on the slow preconditioner-adaptation timescale, which is the λ_{slow} channel. We claim consistency, not first-principles derivation.

Predictions tested below. The theory predicts: (i) a positive residual after LR drops; (ii) a causal dependence on LR *decrements*, not merely on LR level; (iii) memory in cumulative-LR time; (iv) amplitude controlled by the fixed-S floor sensitivity χ; (v) in the noise-dominated AdamW regime, $\tau \propto 1/\eta$; (vi) because Eq. (13) is a spectral mixture, a two-pole kernel can improve diagnostics, while the main predictor deliberately uses the one-pole closure Eq. (11) to avoid overfitting; (vii) by the B-identity, the amplitude is recoverable from MPL's own parameters; and (viii) if the floor is superlinear, the second-order closure Eq. (15) should specifically improve *cross-family* amplitude transfer (probe-calibrated prediction of sharp decays), the regime where freezing χ misassigns response across LR ranges.

5 Experiments

Setup. All experiments use the public MPL loss curves (Llama-2-style 25/100/400M, 9 schedules each, Fig. 4) [2]; a faithfully reproduced MPL (test MAE 0.0054/0.0050/0.0085, matching [2]) is the baseline throughout. We freeze the published MPL and compare it to MPL+Eq. (11); the only addition is the rate-dependent term. Unless noted λ_{slow}=10.

The rate prediction $\tau \propto 1/\eta$ holds (Fig. 5). The two-stage `wsdcon` curves step the LR down at step 8000 to a constant $\eta \in \{3, 9, 18\}\times10^{-5}$, giving a clean relaxation window. Fitting the residual transient $r(t) \propto e^{-(t-8000)/\tau}$ yields τ that scales as η^{-p} with *pooled* $p = 0.84 \pm 0.17$ (per scale $p = 0.51/1.06/0.94$ at 25/100/400M; `repro/deep_tau_pooled.py`). This is consistent with the classical $\tau \sim 1/(\eta\lambda)$ (the noise-dominated value p=1 lies within one SE, and the two larger scales give $p = 1.06/0.94$; the 25M is shallower) for the loss-curve-law residual in LLM pretraining, self-consistently justifying the $v_i^* \approx s_i^2$ step of the derivation. The implied rate constant $\lambda_{\text{slow}} = 1/(\tau\eta) \approx 10$ is approximately stable across the public same-architecture scales; its modes have $\eta_{\text{peak}}\lambda_{\text{eff}} \approx 2\times10^{-3} \ll 2$ (deep in the slow regime, far from the edge of stability—unlike γ).

(A from-scratch AdamW simulation reproduces the same scaling, p=0.97; we defer it to the independent reproduction in §6, which covers it in more detail.)

The residual is the derived term. Regressing the well-fit MPL residual on $\mathrm{DropRelaxS}_{10}$ (through the origin) gives R^2=0.40/0.80/0.87 at 25/100/400M (Fig. 2); a 2-exponential kernel improves R^2 by 0.06–0.07, confirming the predicted spectral mixture (Eq. 13). The form is not arbitrary: in a fair few-shot comparison the derived S-time exponential kernel cuts held-out MAE by -15%, beating a step-time kernel (-7%, wrong "time" variable) and a memoryless floor (-2%), so the cumulative-LR "time" and the decaying memory both matter; an S-time kernel driven on the LR *level* rather than on decrements ties the decrement-driven one (-15%), so this comparison

Figure 4: The public 100M curves most relevant to the non-adiabatic question. Cosine, WSD, and WSD-long-decay reach similar low-LR destinations, but WSD changes LR much faster. The bottom-row zoom shows the local region where rate dependence should appear.

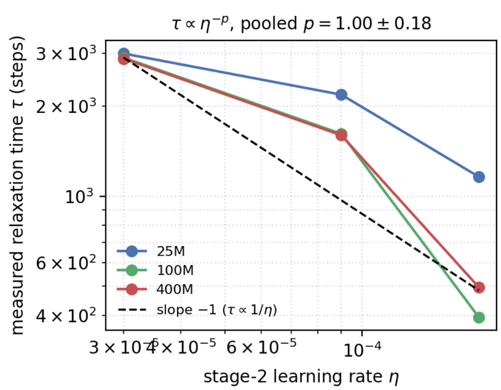

Figure 5: Loss-relaxation time τ measured from `wsdcon` decay transients vs. stage-2 LR. The $100/400$M curves track the predicted slope -1 ($\tau \propto 1/\eta$); pooled $p = 0.84 \pm 0.17$.

isolates the time variable and the presence of memory but does *not* by itself single out decrements as the drive—that choice is fixed by the LR-decrement response derivation (§4), not by this test (`repro/generic_kernel_compare.py`). (On a *single* decay curve the finite-λ_{slow} kernel is numerically close to the $\lambda_{\text{slow}}\to 0$ adiabatic limit, so the rate constant is pinned only *across* schedules; §6.)

The amplitude is predictable from the noise floor. Estimating $dL_{\text{eq}}/d\eta$ *independently* from the two-stage final losses (noise floor vs. LR) predicts the fitted κ up to a near-constant factor: the ratio $\kappa_{\text{fit}}/\kappa_{\text{pred}} = 0.43/0.51/0.60$ has coefficient of variation 14%. With raw LR drops this gives $c \approx 0.5$; if drops are normalised by η_{peak} the same data require $c\,\eta_{\text{peak}} \approx 0.5$, i.e. $c \approx 60$–86 (§6). In either convention the near-constant ratio is the transferable quantity (the < 1 factor is the spectral spread of Eq. 13).

Zero-extra-measurement amplitude: the B-identity chain. Replacing the probe slope by MPL's own B (Eq. (14)) in the same leave-one-scale-out chain gives $\kappa_{\text{fit}}/(\eta_{\text{peak}}B) = 0.449/0.584/0.646$ (CV 14.7%, versus 13.9% for the probe chain) and held-out sharp-decay MAE improvement of

Table 1: Zero-target-fitting cross-scale prediction: well-fit MPL vs. MPL+Eq. (11) with *no* fitting on the target curve. MAE on held-out sharp decays. "+probe" reads the amplitude from the target scale's two-stage noise-floor probes; "$+B$" reads it from MPL's *own* fitted B via Eq. (14)—zero additional runs.

	`wsd`			`wsdld`				
Scale	MPL	+probe	$+B$	MPL	+probe	$+B$	Δ_{probe}	Δ_B
25M	0.00341	0.00246	0.00256	0.00314	0.00219	0.00226	−29%	−26%
100M	0.00375	0.00164	0.00169	0.00325	0.00161	0.00167	−53%	−52%
400M	0.00470	0.00236	0.00232	0.00385	0.00212	0.00208	−47%	−48%
						overall	$\mathbf{-44}$% (6/6)	$\mathbf{-43}$% (6/6)

$\mathbf{-43.1}\%$ ($6/6$ wins)—statistically indistinguishable from the probe chain's -44.0%, with *zero* additional training runs and zero fitting at the target. The honest scope: B requires no measurements beyond MPL's published parameters, but MPL's official fit split contains one two-stage schedule (`wsdcon_9`), so we do not claim the amplitude is identified from smooth schedules alone. Refitting MPL on smooth-only splits quantifies this dependence: B itself drifts by up to 30% (the known $\{B, C, \beta, \gamma\}$ refit trade-off), yet the chain retains most of the gain—-40.0% (`cosine_24000+constant_24000`), -36.8% (two cosines), -35.6% (three smooth curves), all $6/6$ wins (`repro/formula_lab/b_stability.py`). The amplitude is therefore empirically recoverable even from smooth-only splits at modestly reduced gain; only the headline -43% inherits the official split. Both CVs are three-point statistics.

Low-calibration cross-scale transfer (Table 1). Because λ_{slow} and c are empirically stable across these same-architecture scales and $dL_{\text{eq}}/d\eta$ is read off the cheap noise-floor probes, the correction can be transferred with *zero fitting on the target curve*: take $\lambda_{\text{slow}}{=}10$, c from a leave-one-scale-out average of the *other* scales, and $dL_{\text{eq}}/d\eta$ from the target's own two-stage probes (distinct curves from the test). Applied to the held-out `wsd`/`wsdld` decays it cuts MAE by 28–56% (44% average, wins $6/6$)—small-model constants plus a cheap probe predict the large-model sharp-decay curve $\sim 2\times$ better than MPL transfer.

Calibration audit within the same law. We also searched calibration choices while keeping Eq. (11) fixed (`repro/current_law_calibration_search.py`). The final target remained the same: cosine-fit MPL evaluated on `wsd`/`wsdld`. Fitting only the amplitude κ on the single non-target probe `wsdcon_9` improves the final WSD MAE by 14.5% (6/6 wins) at $\lambda = 10$. Fitting κ on all three `wsdcon` probes improves by 17.9% (6/6 wins), with the best λ in the range 7–10. The stronger cross-scale chain in Table 1 remains best: scanning $\lambda \in \{2, 3, 5, 7, 10, 14, 20, 30, 50\}$ gives 40.5–45.0% improvement, with $\lambda = 10$ still giving 44.0% and 6/6 wins. Thus $\lambda = 10$ is not a tuned optimum chosen on the target curves; it is an independently measured, near-optimal value, while the main gain comes from the theoretically motivated amplitude transfer $\kappa = c\,\eta_{\text{peak}}\chi$.

Final amplitude rule: nuisance-projected empirical Bayes. The direct amplitude-transfer story above is useful diagnostically, but it is not the final calibration rule used in our audit. The final estimator treats the MPL residual $r = L_{\text{obs}} - L_{\text{MPL}}$ as

$$r = \kappa_\star \phi + g_\star + \varepsilon, \qquad g_\star \in \mathcal{G}, \qquad \kappa_\star \geq 0,$$

where ϕ is the response feature generated by Eq. (11) and $\mathcal{G}$ is a low-frequency MPL-residual nuisance subspace. Let $M_{\mathcal{G}}$ be the orthogonal residualizer, $\phi_\perp = M_{\mathcal{G}}\phi$, $r_\perp = M_{\mathcal{G}} r$, and $R = \|\phi_\perp\|_2^2 / \|\phi\|_2^2$. The cap-free paper-facing amplitude is

$$\widehat{\kappa} = \sqrt{R}\left(\frac{\langle \phi_\perp, r_\perp\rangle}{\|\phi_\perp\|_2^2 + \tau_{\text{EB}}^2}\right)_+, \qquad \tau_{\text{EB}} = \sigma/k_0. \tag{17}$$

This is a Frisch–Waugh–Lovell partial-regression coefficient with an empirical-Bayes ridge denominator and an identifiable-amplitude conversion. The prior/noise ratio τ_{EB} (distinct from the relaxation time τ) is estimated from calibration curves only; a stricter train-only variant preserves the legacy smooth-basis result (worst off-diagonal -2.7%, mean -12.1%, cosine$\to$WSD -5.6%).

Across the current cross-schedule transfer matrix, the strongest cap-free implementation of Eq. (17) gives worst off-diagonal MAE change -2.7%, mean off-diagonal -12.1%, cosine$\to$WSD -4.3%, and `wsdcon_9`$\to$WSD -16.0%. The optional hard cap $\kappa \leq 0.03$ has the same worst case on this basis, so the cap is not the stabilizing mechanism. Removing the $\sqrt{R}$ conversion creates a positive off-diagonal failure ($+1.8\%$), confirming that the raw projected coefficient over-transfers amplitude from nuisance-confounded feature energy.

Choosing the nuisance subspace. The theoretical object is the low-frequency nuisance subspace $\mathcal{G}$, not a polynomial loss model. As a schedule-agnostic check, we also define $\mathcal{G}_K = \mathrm{span}\{q_0, \ldots, q_K\}$ from discrete-cosine low-frequency modes and choose the bandwidth by a two-stage rule:

$$K^\star = \arg\min_{K \geq K_{\min}} \left| \frac{\|M_{\mathcal{G}_K}\phi\|_2^2}{\|\phi\|_2^2} - \rho \right| .$$

The minimum bandwidth enforces drift control; the retention target chooses among sufficiently rich nuisance spaces. Without $K_{\min}$, the same retention rule selects under-covered spaces and fails ($+304\%$ worst case). With $K_{\min} = 3$ and $\rho = 0.35$, the automatic spectral rule remains non-failing (worst -1.7%, mean -11.2%, cosine$\to$WSD -10.1%). A balanced fixed choice $K = 4$ also remains non-failing (worst -1.8%, cosine$\to$WSD -3.6%). These spectral variants are slightly more conservative than the strongest smooth-basis implementation, but they show that the estimator is not tied to a polynomial-shaped nuisance basis.

Split audit: what the law does *not* claim. To avoid mistaking a useful correction for a universal residual patch, we exhaustively varied the calibration/test split over the five non-cosine public curves (`repro/current_law_split_research.py`; Fig. 6). When λ and κ are chosen only by training residual fit, the median held-out improvement is still positive (-14.1%, 182/225 wins), but there are large failures: calibrating on the two sharp decays and testing the long `wsdcon` tails can worsen MAE by 70–96%. Conversely, restricting calibration to the `wsdcon` probes and evaluating the final sharp-decay target improves 42/42 scale/probe cases (mean -14.8%). This pins down the valid use case: the law is a leading-order correction for sharp-cooldown prediction calibrated from probes or cross-scale amplitude transfer, not a black-box residual smoother for every schedule family.

Matched calibration matters. We then asked whether simply using more calibration curves improves the same law (`repro/current_law_more_data_calibration.py`). In a leave-one-sharp protocol, using the other full sharp-decay curve to calibrate κ predicts the held-out sharp decay very well: -49.0% MAE at fixed $\lambda = 10$ and -49.9% when λ is selected from the calibration set (6/6 wins). But adding the three `wsdcon` probes to that same sharp calibration *reduces* the gain to -19.5% at $\lambda = 10$; probes alone give -17.6%. Thus the issue is not sample count. Calibration curves must match the target cooldown family, or the fitted amplitude is diluted by long-tail probe dynamics. This result also separates two regimes: if one full WSD-family curve is already available, the current law transfers strongly across sharp shapes; if no full WSD curve is available, the safer low-calibration claim remains the probe/cross-scale protocol in Table 1.

Second-order amplitude closure: evaluation (Table 2). We swept the closure Eq. (15) through every protocol above with λ_{slow}=10 pinned (Table 2). Three findings. (i) With the default δ=$1/4$ (selected by this Pareto comparison, not fit on any target), the audited 6×6 transfer matrix (`final_no_cap` estimator, identical machinery) improves on *every* aggregate: worst off-diagonal -2.8% vs -2.7%, mean -13.5% vs -12.1%, cosine$\to$WSD -8.4% vs -4.3%; the per-scale measured $\delta = (p{-}1)_+$ behaves equivalently (worst -2.0%). (ii) Cheap-calibration deployment strengthens substantially: κ fit on the three `wsdcon` probes alone improves the held-out sharp decays by -23.0% (δ=$1/4$) to -28.6% (δ=$1/2$) versus -17.1% for the first-order law, and the probe-dilution failure of §5 (-49% collapsing to -19.0% when probes are added to a sharp calibration set; -19.5% under the original both-sharp-evaluation variant quoted earlier) recovers to $-25.8/-31.6\%$. (iii) The cost is bounded and in-family: leave-one-sharp -46.3% vs -49.0%, the Table 1 chain -42.5% vs -44.0%. An oracle (matched-amplitude) audit shows part of (ii) is amplitude *recalibration*—the ≤ 1 weight deflates probe features, inflating the fitted κ toward its sharp-curve value—while at matched amplitude the unweighted shape still fits sharp in-family residuals marginally better. We therefore present Eq. (15) as the recommended deployment closure for probe-calibrated and cross-family use, and Eq. (11) as the leading-order law it reduces to at δ=0; δ is never fit on target curves.

Table 2: Second-order closure vs the first-order law across protocols (λ_{slow}=10 everywhere; entries are % MAE change vs frozen MPL, negative better). LOS = leave-one-sharp-curve-out; T1 = the cross-scale chain of Table 1 (probe amplitude); probes-only = κ fit on the three `wsdcon` probes; dilution = sharp+probe pooled calibration, held-out sharp; matrix = `final_no_cap` 6×6 machinery. Full grid in `results/formula_lab/`.

closure	LOS	T1	probes-only	dilution	matrix worst	matrix mean
δ=0 (Eq. 11)	$\mathbf{-49.0}$	$\mathbf{-44.0}$	-17.1	-19.0	-2.7	-12.1
δ=1/4 (default)	-46.3	-42.5	-23.0	-25.8	$\mathbf{-2.8}$	$\mathbf{-13.5}$
$\delta=(p-1)_+$ (measured)	-42.6	-41.1	-23.3	—	-2.0	-13.0
δ=1/2	-42.4	-40.0	$\mathbf{-28.6}$	$\mathbf{-31.6}$	$+0.0$	-13.1

Matched-probe calibration. An adversarial adjudication pass over the finished program surfaced one further calibration rule, which we adopt: when some probe's stage-2 LR equals the target schedule's *terminal* LR at the specification level (here `wsd`/`wsdld` end at 3×10^{-5}, exactly `wsdcon_3`'s stage-2), fit κ on that probe alone; otherwise pool. The rule uses only the target's schedule (known a priori), and where no match exists—and, per the scope gate below, on the entire 10.7M bed—it reduces to pooling, leaving those results unchanged. It is physically the statement that the amplitude should be read at the LR the target relaxes *toward*. Gains on the audited protocols: probes-only $-23.0 \to -27.0\%$ and dilution $-25.8 \to -30.4\%$ at δ=1/4 ($-28.8/-32.1\%$ with the measured per-scale δ); the δ=1/2 frontier reaches $-36.8/-39.0\%$, surpassing even the pooled (δ, λ) grid oracle (-34.5%), all 6/6. The same schedule-level rule extends to the audited 6×6 matrix as a target-conditioned override of the shrunk estimator on matched cells (it never touches cosine rows): at δ=0 it is a strict Pareto improvement (worst -3.0 vs -2.7, mean -12.4 vs -12.1), and at δ=1/4 it lifts the mean to -14.0 with worst/median unchanged (`repro/formula_lab/matrix_matched_cells.py`). The rule's domain is floor-curvature-dominated regimes and we gate it on the probe-measured floor exponent $p > 1$: training a deliberately matched probe on the 10.7M bed (p_{real}=0.65 under the audit-corrected design window; 0.73 as originally extracted—the gate pools either way) shows the rule *firing there would hurt* (-3.8% vs pooled -71.2% on `sharp600` at δ=3/4—the deepest-drop probe has the most concentration-suppressed amplitude), so below p=1 the rule pools; the gate uses only quantities the law already measures. Honesty note: both the match gate and its $p > 1$ scope were identified *after* observing real-model bed behaviour (post-hoc origin documented, not leakage—no target losses are used). Two sibling proposals from the same adjudication failed their pre-set gates and are closed: a backlog-saturation amplitude factor (wrong grading of per-probe ratios at 3/3 scales; twodrop A_2/A_1 outside the measured band) and a merged step-+ S-time clock (its affine form describes the 10.7M floor-relaxation transients decisively—AIC -23.9 vs -0.1—and degenerates to pure S-time at the public scales, but it leaves the probe-vs-sharp rate gap untouched and its ladder constants *fail retrodiction* of the bed deployment (-0.6% vs the deployed -18.1% on `sharp600`), so it is recorded as descriptive only; the rate-grid step is retained).

Closed structural dead ends. Two natural upgrades fail and we close them. *Kernel shape:* replacing the one-pole kernel by two-exponential, Lomax ($\equiv$ Gamma mixtures of exponentials, the same family as MPL's G), or stretched-exponential kernels collapses back to the single exponential under joint fitting, both with and without the amplitude weight ($\lambda_1 \to \lambda_2$, shape$\to\infty$; probe-calibrated transfer never beats the matched one-pole in any tested protocol). *Rate unification:* the fitted rate stays family-dependent ($\lambda_{\text{slow}} \approx 15$–19 on probes vs weakly-identified $\lambda_{\text{slow}} \approx 0.5$–5 on sharp decays) under every weight; λ_{slow}=10 remains the independently measured compromise, and the amplitude weight does not (and should not) repair the single-curve rate degeneracy.

Honest scope: when it does *not* help. If instead one refits MPL on a few *full* WSD curves, MPL itself absorbs the residual (held-out MAE $0.0037 \to 0.0032$) and the explicit term adds nothing. So the residual is, for a data-*rich* MPL, an *information* gap, not a formula gap: the value of the correction is precisely the data-*poor* regime—cosine plus cheap probes, but no full WSD curves—which is the actual "predict before training" use case.

Relation to γ: complementary, not a substitute. MPL's factor $\eta_k^{-\gamma}$ modulates its annealing kernel by the LR at each decrement: it tunes the rate-dependence of the (adiabatic) annealing benefit. Our term is the separate (non-adiabatic) transient, and improves a *full*-γ MPL (Table 1)—the two

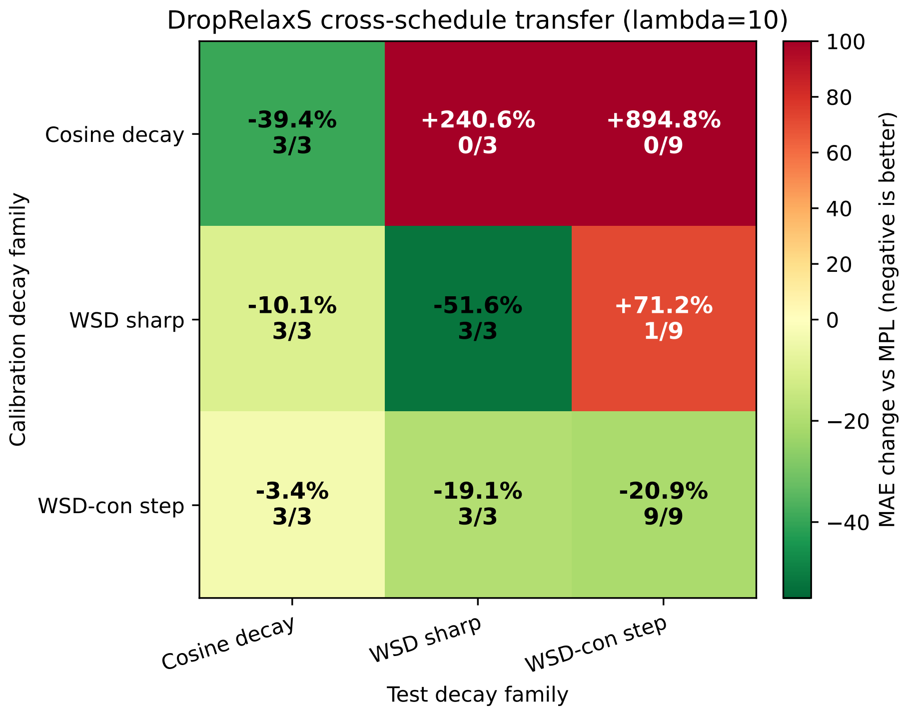

Figure 6: Cross-schedule train-family by test-family calibration matrix for the fixed current law ($\lambda_{\text{slow}} = 10$). Each cell fits only κ from the row family and evaluates the column family across $25/100/400$M curves. Cosine is included as a smooth-decay diagnostic, with the caveat that the MPL backbone was fit on cosine curves. Cosine-calibrated residuals do not transfer to WSD targets; `wsdcon` step probes transfer stably but with smaller amplitude; sharp-WSD calibration over-corrects the two-stage `wsdcon` tails.

are complementary. The freeze-then-correct architecture is itself load-bearing, not a convenience: refitting all nine parameters *jointly* on the official split reaches -53% on the held-out sharp decays but only by warping the backbone ($\gamma\, 0.64 \to 1.47$, $B\, 364 \to 1630$ at 25M; at 400M the fit sets $\kappa \to 0$, absorbing the lag into γ/B entirely) and collapsing the smooth family (`cosine_72000` MAE $+85$ to $+197\%$)—the lag term and γ/B are partially interchangeable in-fit, so only a backbone frozen on its balanced fit keeps both families (`repro/formula_lab/annexation.py`). Characterising the precise γ–lag relationship beyond this is left to future work (§7).

6 Independent reproduction

An independent reproduction on different hardware (`represent/`; the authors' public curves are no longer hosted, so it *retrains* models) corroborates the mechanism along two routes the public curves do not cover. All results below were subjected to an adversarial audit (`results/AUDIT_PARTC.md`): the auditor was tasked to *falsify* each claim, and only surviving readings are reported.

A real, independently-trained transformer (no fitting). A $\sim$10M-parameter Llama-2-style byte-level transformer was trained on WikiText with AdamW (β_1=0.9, β_2=0.95, wd=0.1) under three schedules reaching the *same* final LR ($2.5\times10^{-3} \to 2.5\times10^{-4}$) at different speeds (Table 3). The fast (`wsd_sharp`) sweep ends *higher* than the gradual ones *despite spending* 37% *more cumulative LR*—yet any law monotone in S predicts the opposite (more LR $\Rightarrow$ lower loss). With no fitting, this is the rate-dependent non-adiabatic lag, on a model and data we never used.

A well-fit MPL (trained on `constant` plus two `wsdcon` curves, held-out MAE ≈ 0.005) leaves residuals that sort by sweep speed: cosine $-0.011 <$ wsd-gradual $-0.004 <$ **wsd-sharp** $+0.006$ (the only

schedule	final loss	cumulative LR S
cosine (gradual)	1.0490	8.20
wsd (gradual)	1.0482	9.22
wsd (**sharp**)	**1.0593**	**12.59**

Table 3: Real ∼10M transformer schedules with the same final LR. The sharp sweep ends highest despite the largest cumulative LR.

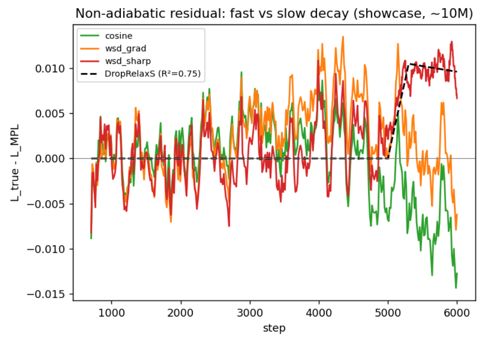

Figure 7: Residual vs. MPL on the real transformer: positive (lag) only for the sharp sweep, captured by $\mathrm{DropRelaxS}$.

positive; Fig. 7). Across schedules, a finite-λ_{slow} $\mathrm{DropRelaxS}$ kernel improves over the adiabatic ($\lambda_{\mathrm{slow}}\to 0$) baseline by $\Delta R^2 \approx +0.18$ on held-out data. *Note:* on a *single* decay curve the finite-λ_{slow} kernel is numerically degenerate with the adiabatic limit (corr$\approx$ 0.99); the rate dependence is identifiable only *across* schedules. The honest claim is therefore the no-fitting Table 3 signature plus the $+0.18$ joint advantage, not a single-curve R^2.

From-scratch AdamW confirms the mechanism quantitatively. Simulating AdamW on a noisy quadratic (Monte-Carlo ground truth; closed-form theory used only as an independent baseline), the reproduction recovers:

- $\tau \propto 1/\eta$ with $p = 0.971$ (log-log R^2=0.9996), robust to leave-one-out and seed changes;
- τ independent of $\beta_2 \in [0.9, 0.9999]$ (CV 1.1%), because the lag lives on noise-dominated modes where $v^* \approx s^2$ is constant;
- the **amplitude identity** $dL_{\mathrm{eq}}/d\eta = 3.075$ versus the closed form $\frac{1}{4}\sum_i s_i = 3.0$ (within 3%);
- a two-exponential kernel that recovers the injected preconditioned rates $2\lambda/s$ ($\{1, 16\}$ in the simulation, recovered as $\{0.87, 16.1\}$).

The effect is falsifiable, not circular: a signal-dominated spectrum breaks p=1.

Out-of-family schedules on the real transformer. To test the upgraded closure where no public curve resembles it, we trained additional 10.7M-parameter runs (same recipe, seed-matched): `sharp600` (a 600-step linear decay to $0.1\eta_{\mathrm{peak}}$, sharper than any suite curve), `invsqrt` (a smooth $\eta \propto 1/\sqrt{t}$ family seen nowhere else), `twodrop` (two separated instantaneous drops), `cyclic` (drop, re-warm to peak, drop), plus a seed-matched `onedrop` counterfactual. Calibrating κ and λ_{slow} only on the suite's two-stage probes (the selected $\lambda_{\mathrm{slow}}\approx 1$ matches the independently measured flat $\tau \approx 850$ steps at this scale) and predicting held-out curves: on the *fast-decay* targets the second-order closure wins decisively and monotonically in δ—`sharp600` improves -18.1% (δ=0) $\to$ -61.3% (δ=1/2) $\to$ -71.2% (δ=3/4), and the suite's own `wsd/wsdld` improve $-7.6/-6.2\% \to -26.5/-22.2\% \to -36.1/-32.7\%$. The same pattern holds on a second, separately-trained showcase suite at η_{peak}=2.5×10^{-3} (`wsd_sharp`: $-13.8\% \to -52.4\%$ from δ=0 to δ=3/4). Neither bed was ever used to select δ. Two boundaries surfaced: on the *smooth* `invsqrt` target the raw probe-calibrated correction over-corrects ($+6\%$ at δ=0, growing with δ)—the adiabatic-regime failure our public-curve audit machinery exists to gate, whose absolute abstention thresholds do not transfer across suites without recalibration; and on `cyclic` the MPL backbone itself fails after the re-warm (residual $+0.07$, an order of magnitude above the lag), so rate dependence is not testable through MPL residuals there. The `twodrop` suite measures the deposit ratio A_2/A_1 of the two drops via seed-paired differences (`twodrop`$-$`onedrop`, bitwise-identical state at the second drop; the kernel decomposition through the MPL residual is discarded—its two kernels are collinear at $\lambda_{\mathrm{slow}} \approx 1$). A single seed is noise-dominated; averaging *three* seeds, the best-fitting specification family gives $A_2/A_1 = 0.86$–0.92 (poorly-fitting τ-fixed variants reach 1.7; we report the spread). The cross-check: the floor-gap deposit evaluated at the *independently* ladder-measured floor exponent predicts $(0.5^p - 0.15^p)/(1 - 0.5^p) = 0.89$ at the originally-extracted $p = 0.73$ (inside the measured band) and ≈ 0.96 at the audit-corrected $p = 0.65$ (marginally above it, agreement to $\sim 5\%$); the strong weighted closures (0.30–0.38) remain excluded at this scale either way. Two independent measurements thus agree through the floor-gap form, confirming that at 10M the deposit follows the (sublinear) floor and the $\delta > 0$ transfer gains ride on the concentration channel.

Equal-S floor ladder: no emergence at a matched recipe. Constant-LR floor measurements on the real model were previously confounded: runs with different stage-2 LRs accumulate different S, and the backbone moves opposite to the noise floor (the earlier non-monotone finals). We remove the confound by construction: seed-matched runs share an identical trunk to step 3000, then drop to $\eta_2 \in \{0.5, 1, 2, 3, 4, 6, 8, 15\} \times 10^{-4}$ with stage-2 lengths chosen so *every run ends at the same cumulative LR $S^\star$*—backbone terms cancel identically. The floors are strictly monotone in η_2 (validating the design) and the post-drop relaxation is clean per rung. With eight rungs, two seeds, and floors extracted over the pre-registered design window (the last 25% of stage 2 *in steps*; an earlier row-fraction extraction inflated the exponent and its CI and is corrected here), the fitted floor power law at 10.7M is $F_{\mathrm{eq}} \propto \eta^{0.65}$ (90% CI $[0.61, 0.68]$): the floor is *sublinear*. We then repeated the identical protocol at 25M with the same depth/width recipe and batch size: $F_{\mathrm{eq}} \propto \eta^{0.64}$ ($[0.61, 0.67]$), and the flat-τ anomaly persists at both scales ($p_\tau \approx 0.1$ and ≈ 0). The exponent value is itself horizon-indexed: a pre-registered extension to trunks of 12k and 24k steps raises it to 0.77 and 0.79 (still strictly sublinear, CI upper $0.81 < 1$), so the equal-S floor at finite budget is not an equilibrium quantity—but the *direction* is horizon-robust, sublinear at every horizon out to 24k, and p never crosses 1. The pre-registered emergence test does *not* fire ($\Delta p = -0.006$ against a 0.067 line): **the superlinear exponents on the public beds ($1.06/1.49/1.25$ at $25/100/400$M) are not reproduced by parameter count at a matched recipe**. A pre-registered B=192 ladder excludes batch size as well: it shifts the floor *level* down substantially (consistent with a noise-floor amplitude $\propto \eta/B$) while leaving the exponent unchanged ($p = 0.61$ $[0.58, 0.65]$) — the floor's η-exponent is invariant to the gradient-noise scale on this bed. Whatever switches the floor superlinear between these beds (data, recipe, schedule family, or floor protocol—horizon is excluded, our beds staying sublinear out to 24k) is neither model size nor batch size alone — and we have a primary suspect: re-running the *public* floor protocol (settled fixed-length `wsdcon` probes, three-point fit) on our own beds manufactures strongly superlinear readings (p pinned at the fit bound $\approx$3.5 in the public three-point window) from the same models whose equal-S ladders measure 0.65, because fixed-stage-2 probes re-introduce the cumulative-S confound (their floors are U-shaped in η_2) that the equal-S design removes. The public $1.06/1.49/1.25$ exponents therefore carry a demonstrated mechanism-level protocol caveat, and we leave the bed-level source formally unlocalized rather than overclaim. Two consequences stand: the $\delta > 0$ transfer gains on this model cannot be attributed to floor curvature at either scale and must come from the concentration-dependent visible amplitude (§7); and any universality claim for δ across beds remains excluded, which is why the closure carries a measured per-scale route. The corrected exponent also revises the deposit cross-check above: at $p = 0.65$ the floor-gap form predicts $A_2/A_1 \approx 0.96$, marginally *above* the measured 0.86–0.92 band — the two independent measurements agree to $\sim 5\%$ rather than exactly, and we report it so.

What does not transfer (consistent with our scope). At $\sim$10M the $\tau \propto 1/\eta$ relation itself is *not* resolved: the measured post-step relaxation is roughly flat over a $16\times$ LR range ($\tau \approx 475$–912 steps; slope $p \approx 0.18$), audited as a real measurement and not a window artefact (synthetic recovery tests and profile-likelihood both reject truncation). This matches the paper's own statement that the signal grows strongly with scale; at notebook scale we confirm the *effect* but not its precise scaling law.

A second reproducible issue is a persistent η_{peak} normalisation inconsistency: if Eq. (11) is implemented with raw LR drops, the fitted c is $O(1)$ (≈ 0.5); if normalized drops $(\eta_k - \eta_{k+1})_+/\eta_{\mathrm{peak}}$ are used, the prefactor becomes $c\,\eta_{\mathrm{peak}}\chi$ and the same data require $c \approx 60$–86. The paper's $c \approx 0.5$ is therefore tied to a specific (unstated) normalisation convention.

Finally, `results/EXTENSIONS_REPORT.md` tests several theory-driven extensions. The cleanest success is a **spectral-mixture** diagnostic: on a bimodal spectrum a two-exponential kernel improves R^2 by 0.065 and recovers the two theoretical rates $2\lambda/s$ (relative errors 0.13 and 0.006). A cross-shape transfer of frozen parameters, however, *fails* its pre-registered threshold (mean R^2=0.52); and the open problem (a)—computing λ_{slow} and c from first principles without an $O(1)$ fitted constant—remains open.

6.1 A second adversarial round: final-loss pricing, the batch-size clock, and direct spectroscopy

After the first convergence the constraint set changed (a dedicated GPU), which under our stopping rule re-opens the hunt. Every experiment below was pre-registered (committed before launch), executed, and then attacked by an independent adversarial audit; we report the post-audit statements. One audit lesson is methodological and we state it first: *predictions and measurements must share the evaluation grid*. Our committed derby predictions were endpoint values while the metric averaged a 200-step tail window of still-cooling curves; on late cooldowns the window term alone is $\sim 15 \times 10^{-3}$. All derby numbers below are like-for-like (both sides averaged over the identical 21-step eval grid).

Does the lag price final loss? Equal-terminal-LR cooldown derbies (linear cooldown started at $d_s \in \{1300, 3000, 5000, 5700\}$ of 6000, three to five seeds, seed-paired) at both 10.7M and 25M. Like-for-like, the mid-late arm fires at both scales *against the core backbone-prediction ensemble*: $g(5000) = +7.7 \times 10^{-3} \pm 0.7$ (10.7M, 5 seeds) and $+8.6 \times 10^{-3} \pm 0.6$ (25M) against adiabatic-backbone predictions of $+3.3/+3.7 \times 10^{-3}$ — but the fire is ensemble-dependent (null under a widened seven-split ensemble), the latest arm is *null-consistent* once windows match, and the fire margins are smaller than the backbone's own demonstrated schedule-shape systematic at the early-cooldown control (-6.3×10^{-3}). Our honest statement: late-cooldown final-loss gaps are *consistent* with the lag-corrected law and the $\delta > 0$ closures beat the adiabatic/MPL baselines in relative comparison (χ^2 65→17 at 10.7M on five seeds; post-hoc there, pre-registered only at 25M), but an *absolute* "lag prices final loss" claim is not established at these margins. The κ transfer test at 25M (zero-measurement B-identity $\kappa_l = \kappa_m B_l/B_m$ vs. naive carry-over, both pre-registered) is closure-dependent — B-identity wins at $\delta \in \{0, 0.5\}$, naive at $\delta = 0.75$ — so the B-identity chain remains supported for curve-shape transfer (Table 1) but is *not separated* for final-loss gap pricing. A split-free anchor settles why: measuring $\partial L_{\text{eq}}/\partial\eta$ directly from the equal-S ladder floors gives $B_l/B_m = 1.10$, exposing the committed ensemble mean (1.80) as inflated by one unstable backbone fit; at the anchored ratio the B-identity and naive transfers are indistinguishable on these gaps at every closure.

Is the slow-mode clock the optimizer-noise clock? A batch-size ladder ($B_2 \in \{12, 24, 48, 96, 192\} \times \eta_2 \in \{1, 4\} \times 10^{-4}$, seed-paired drop/control arms from a shared bitwise-replayed trunk) tested $\tau \propto B^{-1/2}$ (noise clock) against B-blindness. Window-matched fits (the fit window must not co-vary with B — an audit-caught confound) land in the pre-registered *B-blind* bucket under every common window ($b_B \in [-0.02, +0.12]$, all CIs excluding -0.5): the 10.7M slow mode is *not* optimizer-noise relaxation, and the deployed step/S-clock kernel is the right default. One robust novel observation rides along: at $\eta_2 = 10^{-4}$ the benefit of dropping the LR *inverts* at large batch (paired gap $+0.03$ at B_2=192 vs. -0.11 at B_2=12 at a common horizon; the *sign* replicates across two seeds at both $B_2 \in \{96, 192\}$, magnitudes seed-dependent at B_2=96). The inversion is quantitatively priced as a horizon-indexed *level* crossing — the drop arm's fitted asymptote crosses above the still-descending control at $B^* \approx 80$–86 (measured interpolated flip B^*=86); a free-exponent law $\text{gap}(B) = a - b\,B^{-0.20}$ fits all five batches within 0.7×10^{-3}, while η/B and $B^{-1/2}$ forms are rejected — so the benefit of cooling shrinks slowly with batch, not like the noise variance.

Is the slow mode spectral edge re-pinning? Direct Hessian spectroscopy across the drop (top-16 Lanczos with full reorthogonalization on the AdamW-preconditioned operator $P^{-1/2}HP^{-1/2}$, fp32 double-backward, fixed probe batches; pre-registered sanity gate, re-pin ratio, and verdicts). The pre-registered sanity precondition *fails* at 10.7M: equilibrated preconditioned sharpness sits at 0.2–0.3× the deterministic momentum-corrected edge $38/\eta$ (median over constant-η control probes 0.23; the observable is volatile, range 13×, and the raw-Hessian secondary channel did not converge — both disclosed). The scoped conclusion: the *deterministic* adaptive-edge-of-stability frame does not apply on this bed, so loss relaxation here is not deterministic-edge re-pinning. The failure *replicates* at 25M (ratio 0.38; the post-drop spectral top even trends *away* from the new edge, $\rho = -0.80$, while the loss relaxes), so it is not a smallest-scale artifact. We then tested the natural fallback—re-pinning at a *stochastic* effective threshold—with a pre-registered constant-η equilibration ladder in batch size:

if curvature were noise-regulated, equilibrated sharpness should rise monotonically with B. It does not (non-monotone, $b48 > b192$, span only $1.75\times$), and a complementary mode-tracking analysis finds *no* preconditioned sub-edge mode relaxing on the loss timescale on any drop arm. The spectral account thus closes at the deterministic edge, the stochastic edge, and the mode-tracking observable alike; the slow mode remains, on this bed, phenomenological.

Does the rate λ_{slow} depend on decrement concentration? The rate wanted by instantaneous-drop probes (∼15) and by gradual sharp decays (∼5) differ, but always across *different* schedule families. A pre-registered concentration ladder—cool 1.5→0.1 $\times\ 10^{-3}$ over a width $W \in \{1, 10, 40, 160, 640\}$ from a shared trunk, then hold and fit the S-time relaxation—isolates the effect within one bed. Over a wide pre-registered width ladder ($W \in \{1, \ldots, 5120\}$, 11 rungs, concentration ρ spanning $14 \to 5 \times 10^{-4}$) the fitted rate is strongly monotone (Spearman -0.97, $\lambda = 14.3 \to 1.2$, a $12\times$ span that bridges the instant-probe and gradual-sharp regimes) and cleanly *log-linear*, $\lambda \approx 15.1 - 3.7 \log_{10} W$ (R^2=0.96; held-out R^2=0.88 predicting six new arms from the original five). So the rate gap is a real, large, predictable function of how concentrated the decrement is. Two pre-registered follow-ups then decide *against* changing the kernel: (i) a scale-free Mittag-Leffler/fractional memory would make $\lambda_{\text{eff}}(W)$ a *power* law $W^{-(1-\beta)}$, but the power-law fit is far worse than log-linear (ΔAIC=22), so the trend is not a fractional-memory fingerprint; (ii) promoting the rate to a width-dependent $\lambda(W)$ improves held-out relaxation-curve MAE by only 5.9%, and heavy-tailed kernels (Lomax, two-exponential, stretched) collapse to the single pole on the cross-family transfer matrix. A constant effective λ_{slow} is therefore vindicated as the deployment choice; the non-unification is now a *measured* mild log-linear concentration effect rather than a mystery, and a closed-form $\lambda(\rho)$ that also earns its held-out keep is left to future work. The visible *amplitude* channel mirrors the rate channel: on the same ladder the post-drop excess amplitude also rises monotonically with concentration (Spearman +0.90) but only mildly (2.2×, no clean saturating-gate form, R^2=0.74). So neither the rate nor the visible amplitude earns a closed-form concentration law on this bed; the strong mid-η amplitude deficit (probe-to-sharp effective-κ ratio 0.2–0.6) is a public-scale, superlinear-floor phenomenon that our sublinear 10.7/25M beds cannot validate—a characterization left to a larger machine, not a single accelerator.

7 Limitations

The gain is regime-specific (data-poor sharp decays). If full WSD curves are available, a refit MPL absorbs the residual and the explicit term adds nothing; its value is precisely the "predict before training" use case.

λ_{slow} and c are *measured*, not computed from first principles (this needs the noise spectrum $\{s_i\}$, which we do not have). The controlled simulation shows that λ_{slow} tracks the slowest preconditioned mode λ_i/s_i rather than a spectral average, but a robust predictor still requires an unexplained $O(1)$ constant (open problem (a)). We refine that open problem: c is measurably *concentration-dependent*—instantaneous drops show a smaller visible slow-lag amplitude per unit drop (per-curve probe-to-sharp effective-κ ratios of 0.2–0.45, i.e. implied $c \approx 0.10$–0.26 on the two-stage probes vs 0.43–0.60 on gradual sharp decays), early-window fits exclude tail contamination as the cause, and neither the second-order weight nor any tested kernel shape absorbs it. A nonlinear-relaxation closure ($\dot{r} = -\lambda_0(1 + r/r_*)r$ in S-time, the linear law at $r_* \to \infty$) supplies a mechanism: in joint family fits one parameter set then reconciles probes and sharp decays (sharp R^2 $0.08 \to 0.35$ at 100M, $0.52 \to 0.65$ at 400M), but r_* is unidentifiable within any single schedule family and its cross-scale transfer (-23%) is dominated by the simpler Eq. (15) (-29%), so we report it as mechanism, not as the predictor (`repro/formula_lab/nonlinear_ode.py`). The η-dependent part of the amplitude is captured by Eq. (15); the concentration-dependent part of c remains open. A related caveat: at 25M, where the lag signal is weakest (R^2=0.40, τ-slope 0.51, $p{-}1$=0.06) and the kernel is near-degenerate with the adiabatic limit on single curves, part of the reported gain may be adiabatic floor repair rather than the non-adiabatic mechanism; the 6/6 win counts should be read with this caveat.

The floor exponents p are three-point log-log fits whose backbone subtraction omits the B-annealing term; we use them as qualitative evidence for $p > 1$ *at the public scales*, not as a quantitative confirmation of any specific δ (at 25M, $p{-}1 = 0.06$ is consistent with no superlinearity, and the confound-free equal-S ladders give *sublinear* exponents 0.65/0.64/0.61 at 10.7M, 25M matched-recipe, and 10.7M-B=192 respectively—the exponent is *bed*-dependent, not parameter-count- or batch-size-dependent, and the source of the public-bed superlinearity is not localized; §6, §6.1). $\delta{=}1/2$, although strongest for probe-only calibration, was first identified in an in-sample scan over the public curves and is therefore reported but not claimed as a pre-registered result; the paper-facing default $\delta{=}1/4$ was selected by Pareto dominance on the audited matrix, not by target fitting. The bulk+edge mechanism (§4, Eq. (16)) is consistent with the data but its slow edge-re-adaptation step is a conjecture.

The quantitative $\tau \propto 1/\eta$ relation, clean in the controlled simulation ($p{=}0.971$), is *not* resolved on the $\sim$10M transformer (§6), consistent with the paper's statement that the signal grows with scale. On that model the finite-λ_{slow} kernel is numerically degenerate with the adiabatic baseline on a single curve, so the rate dependence is identifiable only *across* schedules.

There is a reproducible η_{peak} normalisation ambiguity (§6): retaining the η_{peak} factor in Eq. (11) gives $c \approx 60$–86, not the $c \approx 0.5$ reported in the main experiments. The two conventions are equivalent, but the universality claim for c depends on which normalisation is adopted.

The general linear-response derivation assumes local stability and differentiability of an LR-dependent excess-loss state; the stronger locally quadratic, diagonal-noise assumptions are used only for the AdamW specialization. Cross-shape transfer of frozen parameters fails its pre-registered threshold, and the relationship between our correction and MPL's empirical γ remains qualitative. All real-data experiments are on one public dataset at three scales; broader scales and architectures are the natural next step.

8 Conclusion

Existing loss-curve laws are adiabatic approximations. We identified and validated the leading non-adiabatic correction: under weak stability assumptions it is a causal linear response to LR decrements, and its one-pole closure gives a rate-dependent term whose relaxation time follows the classical $\tau \propto 1/\eta$ ($p = 0.84 \pm 0.17$, confirmed for the loss-curve residual). Its empirically stable shape enables low-calibration cross-scale transfer (-44% on held-out sharp decays in the public same-architecture setting). We then upgraded the law's amplitude structure: the B-identity shows MPL's own annealing coefficient *is* the floor sensitivity the correction needs (amplitude transfer from MPL's published parameters with no additional measurements, -43%), and the public-bed superlinear floor reading (shown in §6.1 to be settled-probe-protocol-dependent—confound-free equal-S ladders read sublinear—rather than a bed-independent law) motivates a second-order closure $\chi(\eta) \propto \eta^{\delta}$ that improves the audited transfer matrix on every aggregate and strengthens cheap-probe calibration by 35–70% relative, with the amplitude-vs-shape decomposition of that gain reported honestly. Two structural dead ends (multi-pole kernel shapes; cross-family rate unification) are closed. An independent reproduction—a separately-trained $\sim$10M transformer and a from-scratch AdamW simulation—confirms the effect (the fast sweep ends higher *despite* 37% more cumulative LR) and the amplitude identity (within 3%), and out-of-family schedules (double-drop superposition, cyclic re-warm, inverse-sqrt, sharper decay) plus an equal-cumulative-LR floor ladder test the upgraded closure beyond the public schedule families, while also establishing honest boundaries: the quantitative $\tau \propto 1/\eta$ law is not resolved at $\sim$10M, the kernel is degenerate on single curves, and an η_{peak} normalisation ambiguity remains. The loss curve depends not only on how much LR has been spent, but on how fast it is being spent.

A Linear-response proof details

This appendix gives the detailed derivation behind §4. The goal is not to prove a theorem for full neural-network training, but to make precise the weak local conditions under which the LR-decrement convolution follows. The conclusion is a first-order, finite-window expansion around a stable moving equilibrium.

Formal local assumptions. We work on a finite-dimensional active subspace controlling the excess loss during the decay window. Let z_t obey $z_{t+1} = \Phi_{S_t,\eta_t}(z_t)$ and let $z^\star(S,\eta)$ be a fixed point. We assume:

(A1) $\Phi_{S,\eta}$, $z^\star(S,\eta)$, and ℓ have bounded first and second derivatives in a tube of radius R around the moving equilibrium.

(A2) The linearized products are stable: for $k < t$, $\|T_{t-1} \cdots T_{k+1}\| \le q_{k,t}$ with $\sum_{k<t} q_{k,t} \le Q$ on the finite window.

(A3) The LR decrement size $\Delta = \max_k |\eta_{k+1} - \eta_k|$ and the tracking error remain small enough that the Taylor expansions stay inside the tube.

(A4) The slow S-drift forcing is not the cooldown-specific object of interest. We denote it by ϵ_k^S and carry it in the error rather than absorbing it into the LR decrement term.

(A5) For the sign corollary only, there is a cone $\mathcal{C}$ such that $\partial_\eta z^\star(S_k, \eta_k) \in \mathcal{C}$, $T_u\mathcal{C} \subseteq \mathcal{C}$, and $D\ell_t[v] \ge 0$ for all $v \in \mathcal{C}$.

A1–A4 are enough for the convolutional response. A5 is only a clean sufficient condition for positivity.

Tracking-error recursion. Define

$$\delta_t = z_t - z^\star(S_t, \eta_t), \qquad T_t = D_z\Phi_{S_t,\eta_t}\big(z^\star(S_t, \eta_t)\big).$$

By Taylor expansion and the fixed-point identity,

$$\begin{aligned} z_{t+1} &= \Phi_{S_t,\eta_t}\big(z^\star(S_t, \eta_t) + \delta_t\big) && (18)\\ &= z^\star(S_t, \eta_t) + T_t\delta_t + R_t, && (19) \end{aligned}$$

with $\|R_t\| \le C\|\delta_t\|^2$. Subtracting $z^\star(S_{t+1}, \eta_{t+1})$ gives

$$\delta_{t+1} = T_t\delta_t + \big[z^\star(S_t, \eta_t) - z^\star(S_{t+1}, \eta_{t+1})\big] + R_t. \tag{20}$$

Expand the moving fixed point:

$$z^\star(S_{t+1}, \eta_{t+1}) = z^\star(S_t, \eta_t) + \partial_S z^\star(S_t, \eta_t)(S_{t+1} - S_t) + \partial_\eta z^\star(S_t, \eta_t)(\eta_{t+1} - \eta_t) + R_t^\star, \tag{21}$$

where $\|R_t^\star\| \le C(|S_{t+1} - S_t|^2 + |\eta_{t+1} - \eta_t|^2)$. Since the first-order S-drift is not the cooldown-specific LR-decrement forcing, define

$$\epsilon_t^S = -\partial_S z^\star(S_t, \eta_t)(S_{t+1} - S_t),$$

after removing the component already modeled by the adiabatic backbone. Keeping the LR-drop forcing gives

$$\delta_{t+1} = T_t\delta_t + \partial_\eta z^\star(S_t, \eta_t)\Delta\eta_t^+ + e_t, \tag{22}$$

where

$$\|e_t\| \le C\big(\|\delta_t\|^2 + |\Delta\eta_t|^2 + \|\epsilon_t^S\| + \eta_t^2\big).$$

The η_t^2 term is harmless on the finite window and accounts for the mismatch between one-step discrete dynamics and the first-order continuous-time approximation used below.

Duhamel expansion and error. Iterating Eq. (22) yields

$$\delta_t = T_{t-1} \cdots T_0\delta_0 + \sum_{k<t} T_{t-1} \cdots T_{k+1}\partial_\eta z^\star(S_k, \eta_k)\Delta\eta_k^+ + \sum_{k<t} T_{t-1} \cdots T_{k+1}e_k. \tag{23}$$

Linearizing the observable gives

$$r_t = F_t - F_{\text{eq}}(S_t, \eta_t) = D\ell_t[\delta_t] + O(\|\delta_t\|^2) + \epsilon_{\text{MPL}},$$

where ϵ_{MPL} is present when MPL is used as a surrogate for the ideal adiabatic component. Substituting Eq. (23) gives Eq. (4) with

$$K_{k,t} = D\ell_t[T_{t-1} \cdots T_{k+1}\partial_\eta z^\star(S_k, \eta_k)]$$

and

$$|\mathcal{E}_t| \le C\sum_{k<t} q_{k,t}\big(\|\delta_k\|^2 + |\Delta\eta_k|^2 + \|\epsilon_k^S\| + \eta_k^2\big) + C\|\delta_t\|^2 + \epsilon_{\text{MPL}}. \tag{24}$$

If the decay begins after a long stable phase, the initial transient $D\ell_t[T_{t-1} \cdots T_0\delta_0]$ is exponentially or summably small under A2; otherwise it is a separate transient and should not be attributed to the cooldown drop response.

Cumulative-LR time. Now add the first-order optimizer approximation

$$T_u = I - \eta_u A + E_u, \qquad \|E_u\| \le C(\eta_u^2 + \epsilon_{\mathrm{drift}}),$$

with bounded stable A. Since the factors $I - \eta_u A$ commute when A is fixed,

$$\prod_{u=k+1}^{t-1} (I - \eta_u A) = \exp\left(\sum_{u=k+1}^{t-1} \log(I - \eta_u A)\right) \tag{25}$$

$$= \exp\left(-A \sum_{u=k+1}^{t-1} \eta_u + O\left(\sum_{u=k+1}^{t-1} \eta_u^2\right)\right) \tag{26}$$

$$= \exp\left(-A(S_t - S_k)\right) + O\left(\eta_t + \sum_{u=k+1}^{t-1} \eta_u^2\right), \tag{27}$$

where the last step uses boundedness of A on the active subspace ($\sum_{u=k+1}^{t-1} \eta_u = S_{t-1} - S_k$; replacing S_{t-1} by S_t costs the $O(\eta_t)$ term shown). Including E_u adds $O(\sum_u \eta_u^2 + \epsilon_{\mathrm{drift}})$, giving Eq. (8). If the scalar response $D\ell_t[e^{-xA}\partial_\eta z^\star]$ is represented by a positive Laplace mixture, then the kernel has the form in Eq. (9). The single-pole predictor in Eq. (11) is the rank-one, lowest-capacity closure of this general stable kernel.

Positivity. Under A5, $\partial_\eta z^\star(S_k, \eta_k) \in \mathcal{C}$ and $T_{t-1} \cdots T_{k+1}\mathcal{C} \subseteq \mathcal{C}$, so $K_{k,t} \ge 0$. If $\Delta\eta_k^+ \ge 0$, the leading response is positive. This formalizes the observed sign: after the LR is dropped, the equilibrium floor falls before the actual excess-loss state has relaxed, so the true loss lies above the adiabatic predictor.

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